

36) Let A be a 5×3 matrix, let y be a vector in $\mathbb{R}^3$, and let z be a vector in $\mathbb{R}^5$. Suppose $Ay = z$. What fact allows you to conclude that the system $Ax = 4z$ is consistent?

Solution: $Ay = z \Rightarrow 4(Ay) = 4z \Rightarrow A(4y) = 4z$ (By Theorem 5(b))
 $\Rightarrow Ax = 4z$ (By scalar multiplication)

So the vector $x = 4y$ is a solution of the eqn. $Ax = 4z$ $\square$